

FIFTH SEMESTER U.G. DEGREE EXAMINATION, NOVEMBER 2021

(CBCSS—UG)

Mathematics

MTS 5D 01—APPLIED CALCULUS

(2019 Admissions)

Time : Two Hours

Maximum : 60 Marks

Section A

*Answer at least eight questions.**Each question carries 3 marks.**All questions can be attended.**Overall Ceiling 24.*

1. If $g(t) = (t-2)^{\frac{1}{2}}$, find $g(27)$ and $g(5)$?
2. Find the domain and range of $f(x) = \frac{1}{x-3}$.
3. Find $f(g(x))$ where $f(u) = u^2 + 3u + 1$ and $g(x) = x + 1$.
4. Find the distance between the points P $(-2, 5)$ and Q $(4, -1)$.
5. State vertical line test.
6. What is slope of (a) X-axis ; and (b) Y- axis.
7. Find the equation of the line that passes through the points $(3, -2)$ and $(1, 6)$.
8. Find $\lim_{x \rightarrow 1} \frac{x^3 - 8}{x - 2}$.
9. Differentiate the polynomial $y = 5x^3 - 4x^2 + 12x - 8$.

Turn over

10. Find all critical numbers of the function $f(x) = 2x^4 - 4x^2 + 3$.
11. Find all real numbers x that satisfy the given equation $4^{2x-1} = 16$.
12. Evaluate $\log_2 32$.

(8 × 3 = 24 marks)

Section B*Answer at least five questions.**Each question carries 5 marks.**All questions can be attended.**Overall Ceiling 25.*

13. Find :

(a) $\lim_{x \rightarrow \infty} \frac{2x^2 + 3x + 1}{3x^2 - 5x + 2}$.

(b) $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x^2 - 3x + 2}$.

14. For the function $f(x) = 1 - x^2, 0 \leq x < 2$
 $= 2x + 1, x \geq 2$.

evaluate the one sided limits $\lim_{x \rightarrow 2^-} f(x)$ and $\lim_{x \rightarrow 2^+} f(x)$.

15. Find the equation of the tangent line to the curve $y = \sqrt{x}$ at the point where $x = 4$.
16. Find the intervals of increase and decrease for the function $x^2 - 4x + 5$.
17. If $f(x) = 5^{x^2 + 2x}$ find all values of x such that $f(x) = 125$.

18. Discuss the continuity of the function $f(x) = \frac{x+2}{x-3}$ on $-2 < x < 3$ and on the closed interval $-2 \leq x \leq 3$.

19. The Gross Domestic Product (GDP) of a certain country was $N(t) = t^2 + 5t + 106$ billion dollars t years after 1998 :

- (a) At what rate was the GDP changing with respect to time in 2008.
- (b) At what percentage rate was the GDP changing with respect to time in 2008.

(5 × 5 = 25 marks)

Section C

*Answer any **one** question.*

The question carries 11 marks.

20. Graph the function $f(x) = -x^2 + x + 2$. Include all x and y intercepts.

21. A manufacturer's total cost consists of a fixed overhead of \$200 plus production cost of \$50 per unit. Explore the total cost as a function of the number of units produced and draw graph.

(1 × 11 = 11 marks)

FIFTH SEMESTER U.G. DEGREE EXAMINATION, NOVEMBER 2021**(CBCSS—UG)****Mathematics****MTS 5B 09—INTRODUCTION TO GEOMETRY****(2019 Admissions)****(Multiple Choice Questions for SDE Candidates)****Time : 15 Minutes****Total No. of Questions : 15****Maximum : 15 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 15.
2. The candidate should check that the question paper supplied to him/her contains all the 15 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MTS 5B 09—INTRODUCTION TO GEOMETRY

(Multiple Choice Questions for SDE Candidates)

1. Find the equation of the circle with center $(-2, 3)$ and radius 4 :

(A) $x^2 + y^2 + 4x - 6y - 3 = 0.$

(B) $x^2 + y^2 + -3 = 0.$

(C) $x^2 + y^2 + x - 6y - 1 = 0.$

(D) $x^2 + y^2 + -6y - 3 = 0.$

2. Find the center and radius of the circle $(x + 5)^2 + (y - 3)^2 = 36$:

(A) Center $(5, 3)$ and radius 6.(B) Center $(-5, 3)$ and radius 6.(C) Center $(0, 0)$ and radius 7.(D) Center $(-5, 2)$ and radius 5.

3. Does the points $(-2.5, 3.5)$ lie inside, outside or on the circle $x^2 + y^2 = 25$:

(A) $(-2.5, 3.5)$ lies outside the circle. (B) $(-2.5, 3.5)$ lies inside the circle.(C) $(-2.5, 3.5)$ lies on the circle. (D) None of these.

4. Find the length of the latus rectum for $y^2 = 12x$:

(A) 3.

(B) -3.

(C) 12.

(D) -12.

5. Find the length of the latus rectum for $y^2 = -8x$:

(A) -8.

(B) 8.

(C) 12.

(D) -12.

6. Find the axis of the parabola $x^2 = -16y$:

(A) x -axis.(B) y -axis.(C) z -axis.

(D) None of this.

7. Find the equation of directrix of the parabola $y^2 = 10x$:

(A) $y + \frac{5}{2} = 0.$

(B) $x + \frac{5}{2} = 0.$

(C) $x - 5 = 0.$

(D) $y + 5 = 0.$

8. Find the equation of the parabola that satisfies the following conditions: focus (0, 0), directrix $y = 3$.
- (A) $x^2 = -12y$. (B) $y^2 = 25x$.
(C) $x^2 = 25y$. (D) $y^2 = 12x$.
9. Find the co-ordinates of the foci of the ellipse $\frac{x^2}{36} + \frac{y^2}{16} = 1$.
- (A) $(\pm 2\sqrt{5}, 0)$. (B) $(\pm \sqrt{5}, 0)$.
(C) $(\pm 10, 0)$. (D) $(\pm \sqrt{10}, 0)$.
10. Find the co-ordinates of the vertices of the ellipse $\frac{x^2}{4} + \frac{y^2}{25} = 1$.
- (A) $(\pm \sqrt{5}, 0)$. (B) $(0, \pm 5)$.
(C) $(\pm 5, 0)$. (D) $(\pm \sqrt{5}, 0)$.
11. Find the co-ordinates of the foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$.
- (A) $(0, \pm \sqrt{7})$. (B) $(\pm \sqrt{7}, 0)$.
(C) $(0, \pm \sqrt{25})$. (D) $(\pm \sqrt{25}, 0)$.
12. The tangent to an ellipse and making angles 60° with x -axis is ?
- (A) $\sqrt{3}x - y + 7 = 0$. (B) $x + y + 3 = 0$.
(C) $x - 2y = 0$. (D) $2x + y - 4 = 0$.